

# On the Fairness of Additive Welfarist Rules<sup>1</sup>

**Karen Frilya Celine**, Warut Suksompong, Sheung Man Yuen

National University of Singapore

18th Meeting of the Society for Social Choice and Welfare (SSCW)

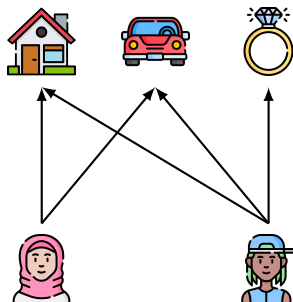
14 June 2026

---

<sup>1</sup>Conference version published at AAMAS 2025. Journal version accepted at TEAC.

# Fair Division

- ▶ We allocate **resources** among a group of agents.
  - ▶ Example: **inheritance**.
- ▶ How to ensure **fairness** and **efficiency**?



*Credit: icons by Freepik*

# Definitions

- ▶ An **instance** consists of:
  - ▶ A set of **agents**  $N = \{1, \dots, n\}$ .
  - ▶ A set of **indivisible goods**  $M = \{g_1, \dots, g_m\}$ .
  - ▶ Each agent  $i \in N$  has a **utility** function  $u_i$ , which is **non-negative** and **additive**.
- ▶ An **allocation**  $(A_1, \dots, A_n)$  is an ordered partition of the goods  $M$ ,
  - ▶ where each agent  $i \in N$  is allocated **bundle**  $A_i$ , which may be empty.

# Definitions

## ► Fairness

- **Envy-free up to one good (EF1)**: For all agents  $i, j$ , either  $A_j = \emptyset$  or there is a good  $g \in A_j$  with

$$u_i(A_i) \geq u_i(A_j \setminus \{g\}).$$

## ► Efficiency

- **Additive welfarist rule** with a strictly increasing function  $f$ :  
Chooses an allocation that maximizes

$$\sum_{i=1}^n f(u_i(A_i)).$$

- Additive welfarist rules guarantee **Pareto optimality (PO)**, i.e.:
  - no other allocation makes some agent better off and no agent worse off.
- Examples:
  - **Maximum utilitarian welfare (MUW)**:  $f(x) = x$ .
  - **Maximum Nash welfare (MNW)**:  $f(x) = \log x$ .
  - **Maximum harmonic welfare (MHW)**:  $f(x) = \sum_{k=1}^x 1/k$ .

# Past Work

## ► Real-valued:

- **MNW** guarantees EF1 [Caragiannis, Kurokawa, Moulin, Procaccia, Shah, and Wang, 2019].
- **MNW** is the only additive welfarist rule that guarantees EF1 for all real-valued instances [Suksompong, 2023].

## ► Integer-valued:

- **MHW** guarantees EF1 for all integer-valued instances [Montanari, Schmidt-Kraepelin, Suksompong, and Teh, 2025].
- An important subclass in terms of real world application.
  - Example: **Spliddit** (<http://www.spliddit.org/>).

## Question

For each subclass of instances (e.g. integer-valued, two-value), which additive welfarist rules guarantee EF1?

# Motivating Example

|           | $g_1$ | $g_2$ | $g_3$    | $\dots$  | $g_n$ |
|-----------|-------|-------|----------|----------|-------|
| Agent 1   | $n$   |       |          |          |       |
| Agent 2   | $n-1$ | 1     |          |          |       |
| Agent 3   |       | $n-1$ | 1        |          |       |
| $\vdots$  |       |       | $\ddots$ | $\ddots$ |       |
| Agent $n$ |       |       |          | $n-1$    | 1     |

**MNW:** Agent 1 gets utility  $n$  and every other agent gets utility 1.

**MHW:** Agent 2 gets utility 0 and every other agent gets utility  $\geq n-1$ .

MHW allocation is arguably **better** than MNW allocation when  $n$  is large.

# Motivating Example

|           | $g_1$ | $g_2$ | $g_3$    | $\dots$  | $g_n$ |
|-----------|-------|-------|----------|----------|-------|
| Agent 1   | $n$   |       |          |          |       |
| Agent 2   | $n-1$ | 1     |          |          |       |
| Agent 3   |       | $n-1$ | 1        |          |       |
| $\vdots$  |       |       | $\ddots$ | $\ddots$ |       |
| Agent $n$ |       |       |          | $n-1$    | 1     |

**MNW:** Agent 1 gets utility  $n$  and every other agent gets utility 1.

**MHW:** Agent 2 gets utility 0 and every other agent gets utility  $\geq n-1$ .

MHW allocation is arguably **better** than MNW allocation when  $n$  is large.

# Motivating Example

|           | $g_1$ | $g_2$ | $g_3$    | $\dots$  | $g_n$ |
|-----------|-------|-------|----------|----------|-------|
| Agent 1   | $n$   |       |          |          |       |
| Agent 2   | $n-1$ | 1     |          |          |       |
| Agent 3   |       | $n-1$ | 1        |          |       |
| $\vdots$  |       |       | $\ddots$ | $\ddots$ |       |
| Agent $n$ |       |       |          | $n-1$    | 1     |

**MNW:** Agent 1 gets utility  $n$  and every other agent gets utility 1.

**MHW:** Agent 2 gets utility 0 and every other agent gets utility  $\geq n-1$ .

MHW allocation is arguably **better** than MNW allocation when  $n$  is large.



# Our Contribution

## ► Real-valued:

- **No restriction:** MNW is the only additive welfarist rule ensuring EF1 [Caragiannis, Kurokawa, Moulin, Procaccia, Shah, and Wang, 2019; Suksompong, 2023].
- The same holds even in more restricted subclasses:
  - **identical-good** instances and
  - **two-value** instances.

## ► Integer-valued:

- There are infinitely many other additive welfarist rules ensuring EF1.
  - Example:  $f(x) = \log(x + c)$  for any  $c \in [0, 1]$ .
- We give necessary and sufficient conditions.

## ► Binary:

- An additive welfarist rule with function  $f$  guarantees EF1 if and only if  $f$  is **strictly concave on  $\mathbb{Z}_{\geq 0}$** .
  - Example:  $f(x) = \text{sgn}(p) \cdot 1/x^p$  ( **$p$ -mean rule**) for any  $p < 1$ .

# Identical-Good Instances

An instance is **identical-good** if

- ▶ for each agent  $i$ , there exists  $a_i \in \mathbb{R}_{>0}$  such that  $u_i(g) = a_i$  for every good  $g$ .

## Theorem 1

Let  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R} \cup \{-\infty\}$  be strictly increasing on  $\mathbb{R}_{\geq 0}$  and continuous on  $\mathbb{R}_{>0}$ . The following are equivalent:

1. An additive welfarist rule with function  $f$  **guarantees EF1 for all identical-good instances.**
2. There exist constants  $\alpha \in \mathbb{R}_{>0}$  and  $\beta \in \mathbb{R}$  such that  $f(x) = \alpha \log x + \beta$  for all  $x \in \mathbb{R}_{\geq 0}$ .

# Proof of Theorem 1

An additive welfarist rule with function  $f$  guarantees EF1 for all identical-good instances.

Lemma 2

$$f((k+1)b) - f(kb) > f((k+2)a) - f((k+1)a) \\ \text{for all } k \in \mathbb{Z}_{\geq 0} \text{ and } a, b \in \mathbb{R}_{>0}.$$

Lemma 3

The function  $\Delta_{f,k}(x) = f((k+1)x) - f(kx)$  is constant on the domain  $\mathbb{R}_{>0}$  for all  $k \in \mathbb{Z}_{\geq 0}$ .

[Suksompong, 2023]\*

There exist constants  $\alpha \in \mathbb{R}_{>0}$  and  $\beta \in \mathbb{R}$  such that  $f(x) = \alpha \log x + \beta$  for all  $x \in \mathbb{R}_{\geq 0}$ .

[Caragiannis, Kurokawa, Moulin, Procaccia, Shah, and Wang, 2019]

## Lemma 2

Let  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R} \cup \{-\infty\}$  be strictly increasing on  $\mathbb{R}_{\geq 0}$  and continuous on  $\mathbb{R}_{>0}$ . If an additive welfarist rule with function  $f$  guarantees EF1 for all identical-good instances, then for all  $k \in \mathbb{Z}_{\geq 0}$  and  $a, b \in \mathbb{R}_{>0}$ ,

$$f((k+1)b) - f(kb) > f((k+2)a) - f((k+1)a).$$

**Proof:** Suppose otherwise. Then, for some  $k \in \mathbb{Z}_{\geq 0}$  and  $a, b \in \mathbb{R}_{>0}$ ,

$$f((k+1)a) + f((k+1)b) \leq f((k+2)a) + f(kb).$$

We construct an instance with  $n$  agents and  $(k+1)n$  goods such that for each good  $g$ ,  $u_1(g) = a$  and for each agent  $i \in N \setminus \{1\}$ ,  $u_i(g) = b$ .

- ▶ **Allocation 1 (unique EF1 allocation):** Every agent receives  $k+1$  goods.
  - ▶  $\sum_{i \in N} f(u_i(A_i)) = f((k+1)a) + (n-1) \cdot f((k+1)b).$
- ▶ **Allocation 2 (not EF1):** Agent 1 receives  $k+2$  goods, agent 2 receives  $k$  goods, and every other agent receives  $k+1$  goods.
  - ▶  $\sum_{i \in N} f(u_i(B_i)) = f((k+2)a) + f(kb) + (n-2) \cdot f((k+1)b).$

### Lemma 3

Let  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R} \cup \{-\infty\}$  be strictly increasing on  $\mathbb{R}_{\geq 0}$  and continuous on  $\mathbb{R}_{> 0}$ . If for all  $k \in \mathbb{Z}_{\geq 0}$  and  $a, b \in \mathbb{R}_{> 0}$ ,

$$f((k+1)b) - f(kb) > f((k+2)a) - f((k+1)a),$$

then the function

$$\Delta_{f,k}(x) = f((k+1)x) - f(kx)$$

is **constant** on the domain  $\mathbb{R}_{> 0}$  for all  $k \in \mathbb{Z}_{\geq 0}$ .

**Observations:** For any  $k \in \mathbb{Z}_{> 0}$ :

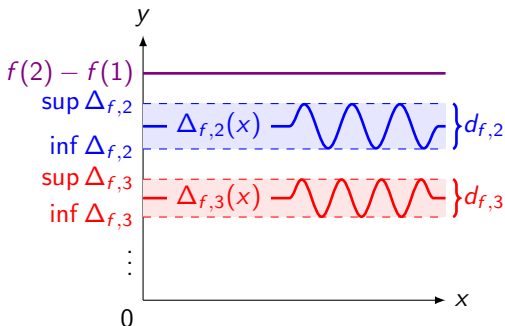
$$1. \Delta_{f,k}(x) > 0 \text{ for all } x \in \mathbb{R}_{> 0}.$$

$$2. \Delta_{f,k}(b) > \Delta_{f,k+1}(a) \text{ for all } a, b \in \mathbb{R}_{> 0}.$$

$$3. \inf \Delta_{f,k} \geq \sup \Delta_{f,k+1} > 0.$$

$$\text{Let } d_{f,k} = \sup \Delta_{f,k} - \inf \Delta_{f,k} \geq 0.$$

$$4. \sum_{t=2}^k d_{f,t} \leq f(2) - f(1).$$



### Proof Sketch of Lemma 3 (Cont'd):

$$\begin{aligned}\Delta_{f,k}(x) &= f((k+1)x) - f(kx) \\ &= f\left((2k+2)\frac{x}{2}\right) - f\left((2k)\frac{x}{2}\right) \\ &= f\left((2k+2)\frac{x}{2}\right) - f\left((2k+1)\frac{x}{2}\right) + f\left((2k+1)\frac{x}{2}\right) - f\left((2k)\frac{x}{2}\right) \\ &= \Delta_{f,2k+1}\left(\frac{x}{2}\right) + \Delta_{f,2k}\left(\frac{x}{2}\right).\end{aligned}$$

Hence,

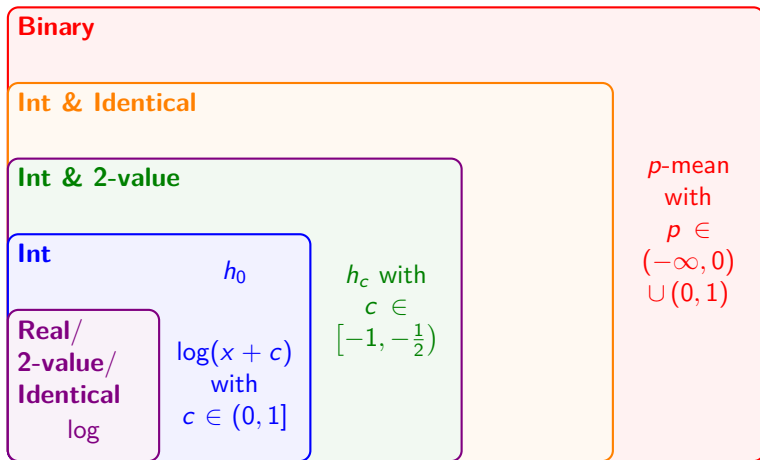
$$\begin{aligned}d_{f,k} &\leq d_{f,2k+1} + d_{f,2k} \\ d_{f,k} &\leq d_{f,4k+3} + d_{f,4k+2} + d_{f,4k+1} + d_{f,4k} \\ &\vdots \\ + \quad d_{f,k} &\leq d_{f,2^c k + 2^c - 1} + d_{f,2^c k + 2^c - 2} + \cdots + d_{f,2^c k + 1} + d_{f,2^c k}\end{aligned}$$

---

$$\begin{aligned}c d_{f,k} &\leq \sum_{t=2}^{2^c k + 2^c - 1} d_{f,t} \\ \Leftrightarrow d_{f,k} &\leq \frac{1}{c} \sum_{t=2}^{2^c k + 2^c - 1} d_{f,t} \leq \frac{1}{c} (f(2) - f(1)).\end{aligned}$$

Since this is true for any  $c \in \mathbb{Z}_{>0}$ , we have  $d_{f,k} \leq 0$ .

# Summary and Examples



Note:  $h_c(x) = \sum_{t=1}^x \frac{1}{t+c}$  is the “modified harmonic number”.

# Future Work

## ► Other settings:

- Non-additive welfarist rules
- Other fairness notions (e.g. PROP1)

## ► Comparison:

- For some subclasses (e.g. integer-valued instances), there are many additive welfarist rules that yield EF1.
- Which rule is the “best”?
- Method of comparison: e.g. price of fairness [Bei, Lu, Manurangsi, and Suksompong, 2021; Celine, Dzulfikar, and Koswara, 2023; Li, Liu, Lu, Tao, and Tao, 2024].

- **Computation:** How to find allocations returned by additive welfarist rules other than MNW?